SYAMA | SAG PEBBLE (SCAT) ORE SORTING

Assessment of the business case and recommendation on STEINERT ore sorting

Field	Detail	Field	Detail
Date	5 August 2026	**Basis**	STEINERT model SR241558 (Sukari), re-parameterised
Subject	Sorting of SAG scats, converted (ex-oxide) line	**Key input**	**Scat grade 2.0 g/t Au (client sampling)**

1. Recommendation

Conditional GO — fund test work, not the plant. The re-parameterised case returns an NPV of US$33.6 M with a 6.5-month payback on US$4.1 M of capital, but the entire result rests on one assumption that has not been tested: that freed SAG capacity gets refilled with ore. Strip that away and the same plant destroys US$13.9 M. Two gates must be closed before any capital is committed.

- Gate 1 (commercial, no cost, ~4 weeks) — confirm that comminution, not the roaster/flotation circuit or ore supply, is the binding constraint on the converted line post-SSCP, and that ore exists to fill it.
- Gate 2 (technical, ~US$60–150 k, ~3 months) — a bulk XRT/sensor test on 2–5 t of Syama scats. A 1.8× mass contrast is not proof of a detectable sensor contrast.

Do not order a sorter until both gates pass. The downside case is not marginal — it is a materially negative NPV plus permanent gold loss to the waste dump.

2. What the work to date establishes

The Sukari SR241558 model simulates a pebble sorter on the SAG mill discharge screen oversize. Barren scat mass is rejected, the circulating load falls, freed SAG capacity is refilled with ROM, and mill/leach feed tonnage is held constant by Goal Seek so the benefit reports as higher feed grade. At Sukari it produced +7,925 oz/a for US$8.2 M. The engine is sound and transfers directly to Syama; what does not transfer is the material property that made Sukari work.

Parameter	Sukari	Syama	Why it matters
ROM / scat grade (g/t Au)	1.57 / 0.56	**2.40 / 2.00**	Sukari scats were barren; Syama scats are not
Scat grade as % of ROM	35%	**83%**	Gold is not rejecting to the coarse fraction
Heterogeneity contrast	8.9×	**1.8×**	Back-solved waste:ore deportment to scats — the crux
Barren mass available in scats	~69%	~33%	Hard ceiling on what any sorter can reject
Reject grade achieved (g/t)	0.148	**0.429**	Gold walking out to the dump

3. Assessment — what a 2 g/t scat grade means

- The scats are not a waste stream. At 2.0 g/t they run at 83% of ROM grade and carry 12.5% of the gold in the circuit. Back-solving the deportment against 20% SLC dilution gives a waste:ore contrast to scats of only 1.8× — against 8.9× at Sukari. Roughly one third of the scat stream is barren dilution; that is the hard ceiling on mass rejection.
- Every rejected tonne is expensive. At 0.43 g/t and US$4,000/oz the reject stream holds US$41/t of recoverable gold against US$10.50/t of milling and handling cost avoided — a 3.9× ratio. The sorter must be tuned for high ore recovery (≥95%), which caps mass pull at ~25% and forfeits roughly a third of the theoretical prize.
- The comminution problem is already being solved. The SSCP installs a secondary crusher and a pebble crusher specifically to handle scats. Crushing recovers 100% of the contained gold; sorting discards some of it. Sorting must beat crushing, not merely beat doing nothing.
- Nothing here is a recovery gain. Scats recirculate — no gold is currently being lost. The value is purely throughput debottlenecking, which is worth exactly nothing if the SAG is not the binding constraint.

4. Economics — and the single factor that decides them

Scenario (realistic XRT: 95% ore rec / 70% waste rej)	Extra oz/a	Net US$/a	NPV @10%
Freed capacity fully refilled with 2.4 g/t ore	+2,699	+7.58 M	**+33.6 M**
Half refilled	+1,066	+2.81 M	+9.9 M
NPV breakeven	+388	+0.82 M	**0 (29% capture)**
No spare ore — ROM held flat	−567	−1.96 M	**−13.9 M**
Refilled with 1.2 g/t stockpile instead	+1,066	+1.05 M	+1.1 M

Secondary sensitivities are mild by comparison: at US$2,500/oz the case still returns +US$13.4 M NPV, and pushing the sorter to 90/90 adds only US$6 M. The scat grade itself is the dominant material lever — at 0.8 g/t the same plant would be worth US$76 M. At 2.0 g/t, roughly 60% of the Sukari-style prize has already been lost to the ore's own character.

If the scat grade were...	Contrast	Reject g/t	Mass pull	NPV @10%
0.8 g/t (Sukari-like)	7.5×	0.10	48%	+76.4 M
1.2 g/t	4.4×	0.18	40%	+63.2 M
1.6 g/t	2.9×	0.28	32%	+49.0 M
2.0 g/t — Syama, as measured	**1.8×**	**0.43**	**25%**	**+33.6 M**
2.4 g/t (= ROM grade)	1.0×	0.72	17%	+16.8 M

Read down that table: the scat grade alone has already removed more than half the prize before a single sorter is specified, and it does so by making the reject stream progressively more valuable. Sorter performance is the weaker lever — running 90% rejection at 90% ore recovery instead of 70/95 raises NPV only to US$39.6 M, and pushing harder (95/85) turns back down to US$37.7 M because the gold lost outruns the extra mass rejected. There is no sorter setting that rescues a poor contrast.

5. Why Gate 1 is the real risk

Public disclosure points away from comminution being the constraint. Sulphide stockpiles were drawn down through 2025 to maintain throughput — a mill short of ore, not short of capacity. The SSCP is adding 60% of sulphide capacity (2.4 → 4.0 Mtpa), and the declared debottlenecking target is the roaster and its new ESP, not the mills. If flotation, the roaster or ore supply binds first, every tonne of freed SAG capacity is worth zero and the case collapses to the −US$13.9 M row above. This must be settled from the plant's own constraint data before anything else is spent.

6. Risks and adjacent opportunity

- Refractory gold — reject grades must be confirmed by fire assay and diagnostic leach, not sensor response. Gold is locked in sulphides and XRT sees density, not gold.
- Sulphidic rejects — geochemical/ARD characterisation is required before a new reject stream is sent to a waste dump; this can carry its own closure cost.
- Mali country risk — supply chain and spares availability disrupted through 2026; a 10% discount rate is applied and single-sorter redundancy is nil.
- Larger prize elsewhere: SLC dilution rejection on ROM or coarse-crushed underground feed. Some 20% dilution across ~2.6 Mt/a offers far more heterogeneity than the scats do, and the same sample campaign can test it at marginal extra cost. Recommend it be scoped into Gate 2.

7. Next steps

#	Action	Owner	Cost	Duration
1	Constraint audit of the converted line post-SSCP; confirm spare ore at reserve grade	Site metallurgy / planning	nil	4 weeks
2	Survey and assay the scat stream — tonnage, size distribution, grade by size, lithology split	Site metallurgy	<$20 k	4 weeks
3	Bulk XRT/sensor amenability test, 2–5 t, scats + coarse ROM	STEINERT test centre	$60–150 k	3 months
4	ARD / geochemical characterisation of the reject stream	Environmental	<$30 k	2 months
5	Re-run this model on measured sorter response; decide on capital	Study team	nil	2 weeks

Model: Syama_Pebble_Sorting_Simulation.xlsx (Dashboard · MB Base · MB Sort · NPV · Sensitivity). All costs in US$. Figures flagged "Assumption – CONFIRM" on the Dashboard are placeholders pending site data: SAG line feed rate 200 t/h, SLC dilution 20%, scat load 30 t/h, milling cost US$8.00/t and incremental mining cost US$53.00/t (derived from AISC, not from site costing).